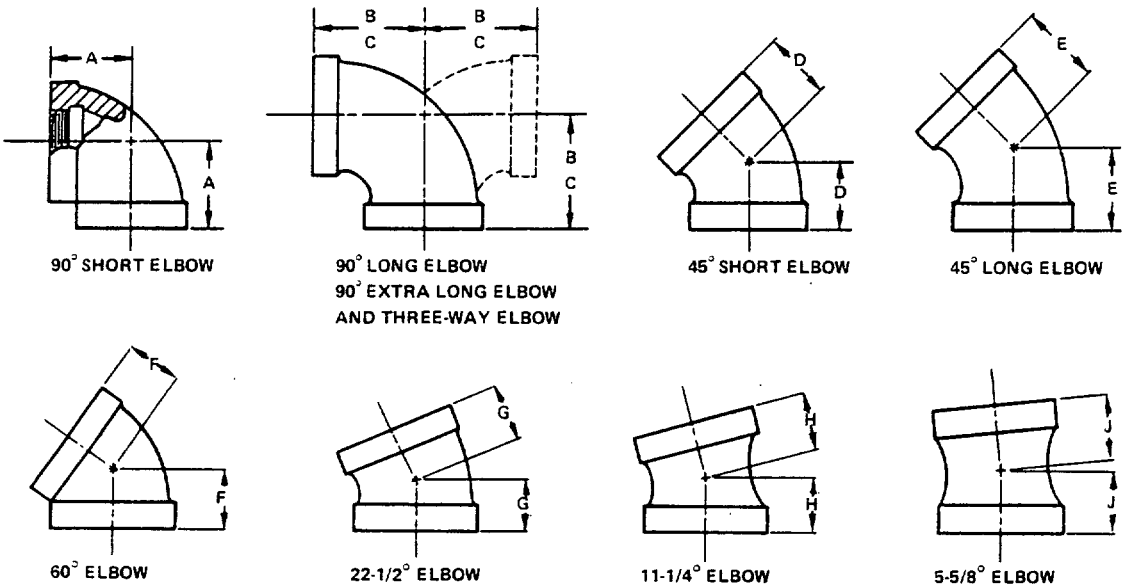
FIGURE 1



Ⓐ TABLE I. Center To End Dimensions.

Nominal Pipe Size	90 Deg. Elbow			45 Deg. Elbow		60 Deg Elbow	22-1/2 Deg Elbow	11-1/4 Deg Elbow	5-5/8 Deg Elbow
	Short	Long and 3-Way Elbow	Extra Long	Short	Long				
	A	B	C	D	E	F	G	H	J
1-1/4	1-3/4	2-1/4	3	1-5/16	1-3/4	1-9/16	1-1/8	1-1/16	1-3/16
1-1/2	1-15/16	2-1/2	3-1/2	1-7/16	1-7/8	1-3/4	1-1/4	1-1/4	1-5/16
2	2-1/4	3-1/16	4	1-11/16	2-1/4	2-1/16	1-7/16	1-3/8	1-1/2
2-1/2	2-11/16	3-11/16	4-1/2	1-15/16	2-5/8	2-1/2	1-3/4	1-5/8	1-5/8
3	3-1/16	4-1/4	5-1/4	2-3/16	2-15/16	2-7/8	2	1-13/16	1-3/4
4	3-13/16	5-3/16	6-1/4	2-5/8	3-1/2	3-3/8	2-5/16	2	1-7/8
5	4-1/2	6-1/8	...	3-1/16	4-1/8	3-7/8	2-5/8	2-1/4	2
6	5-1/8	7-1/8	...	3-7/16	4-7/8	4-1/4	2-15/16	2-3/8	2-1/4
8	6-9/16	9	...	4-1/4	...	...	...	...	...

Ⓐ NOTES:

1. MATERIAL: Material shall be in accordance with ANSI B16.12.
2. COATINGS: Coatings shall be in accordance with ANSI B16.12 and shall be uncoated, coated with a bituminous composition or zinc coated as specified by the MS part number.
3. DIMENSIONS: All dimensions are in inches and shall be in accordance with ANSI B16.12. Three-way elbows have the same dimensions as 90 degree long radius elbows.
4. THREADED ENDS: The threaded ends shall be in accordance with ANSI B16.12. All drainage fittings having openings at 90 degree nominal angle shall be tapped with a pitch of 1/4 inch in one foot and with the pitch in the direction shown in ANSI B16.12.
5. GENERAL:
   (a) Referenced documents shall be of the issue in effect on date of invitations for bid or requests for proposal.
   (b) For design feature purposes, this standard takes precedence over procurement documents referenced herein.

Ⓐ DENOTES CHANGES

Ⓐ User activities: Army - CE
Navy - MC
Air Force - 99

Ⓐ Review activities: Army -
Navy -
Air Force -
DLA-CS

Ⓐ REVISED 11 January 1983
APPROVED 31 October 1972

P.A NAVY-YD Other Cust ARMY-ME Ⓐ	International Interest	TITLE ELBOWS, CAST-IRON THREADED DRAINAGE	MILITARY STANDARD MS90522
Procurement Specification ANSI B16.12 Ⓐ		SUPERSEDES:	PAGE 1 OF 2

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6. PART NUMBER: The MS part number is formed by combining the MS numbers with a dash after it, the size code number, the elbow style letter, and the coating identification number.

MS 90522 - X X X

Size code number (see note 7)
Elbow style letter (see note 8)
Coating identification number (see note 9)

7. SIZE: The nominal pipe size is designated by a digit (see table II).

TABLE II. Code Number to Size.

Code Number	Nominal Pipe Size	Code Number	Nominal Pipe Size
1	1 1/4 inch	6	4 inch
2	1 1/2 inch	7	5 inch
3	2 inch	8	6 inch
4	2 1/2 inch	9	8 inch
5	3 inch		

8. ELBOW STYLE: The elbow style is designated by a single letter (see table III).

TABLE III. Code Letter to Elbow Style.

Code Letter	Elbow Style	Code Letter	Elbow Style
K	90 degree short	R	45 degree long
L	90 degree long	T	60 degree
M	90 degree extra long	W	22 1/2 degree
N	three-way	X	11 1/4 degree
P	45 degree short	Y	5 5/8 degree

9. COATING: The type of coating is designated by a digit (see table IV).

TABLE IV. Code Number to Coating.

Code Number	Coating
1	Uncoated
2	Bituminous composition
3	Zinc coated

User activities: Army - CE
Navy - MC
Air Force - 99

Review activities: Army -
Navy -
Air Force -
DLA-CS

APPROVED 31 October 1972 REVISED (A) 11 January 1983

P.A	International interest	TITLE	MILITARY STANDARD
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Procurement Specification ANSI B16.12 (A)		SUPERSEDES:	PAGE 2 OF 2

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